

Exam Harmonization Reference

Paper 3 — "Beaches or Books": Tourism Sector Shocks and Household Investment in Human Capital

Document purpose: Reference for variable construction decisions when writing the data section of Paper 3. Covers the two harmonized exam attainment variables derived from

`SCHOOL_HIGHEST_EXAM_PASSED` in the JSLC education module.

Script:

`HumanCapital_Tasks/Cleaning_JSLC_2/Code/cleaning_jslc2_educsection2_harmonize.R`

Background: The JSLC Exam Variable

The JSLC education module asks respondents (or their household members) to report the highest educational qualification they have attained. This is stored in

`SCHOOL_HIGHEST_EXAM_PASSED` after column renaming. The variable is problematic for panel analysis for three reasons:

1. **Label drift across waves.** The same credential is described with different strings across years (e.g., "CXC Gen, / GCE O" in 2006 vs. "CSEC General/GCE O Level" in 2014 vs. "CSEC Gen,/GCE O LEVEL" in 2017). This reflects both the real-world rebranding of CXC exams over time and inconsistent SPSS label management by STATIN.
 2. **Numeric code leakage.** In 2015 and 2019, `haven::as_factor()` was not applied during cleaning, so the variable contains raw integer codes (1-13 in 2015; 01-21 in 2019) rather than string labels.
 3. **Category evolution.** The survey instrument itself changed between waves. NVQJ credentials (National Vocational Qualification of Jamaica) were not listed as a response option before 2012. From 2016 onward, NVQJ levels were explicitly merged with their academic equivalents in composite categories (e.g., "TERTIARY CERT./DIPLOMA/ NVQJ LEVEL 3").
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Step 1: `harmonize_highest_exam()` → `SCHOOL_HIGHEST_EXAM_HARMONIZED`

The first function collapses all raw label variants and numeric codes into 15 canonical strings. It uses `grepl()` with `ignore.case = TRUE` for pattern-based matching, which handles minor spacing and punctuation differences automatically. This also means that sub-variants of CAPE (e.g., "CAPE Unit 1", "CAPE Unit 2/GCE A Level", "CAPE UNIT 1 and 2") are all caught by a single

`grepl("cape", ...)` pattern and collapsed to the single canonical string `"CAPE/GCE A Level"` before any downstream recoding.

Canonical output strings from `SCHOOL_HIGHEST_EXAM_HARMONIZED`

Canonical String	Credentials Collapsed
<code>"None"</code>	NONE, None, none, code 1 (2015), code 01 (2019)
<code>"Junior High School Certificate"</code>	JUNIOR HIGH SCHOOL CERT, JUNIOR HIGH SCHOOL CERTIFICATE, codes 2 (2015), 02 (2019)
<code>"Grade Nine Achievement Test"</code>	GRADE NINE ACHIEVEMENT TEST, Grade Nine achievement Test, Grade 9 achievement test, codes 3 (2015), 03 (2019)
<code>"CSEC Basic/JSC 5/SSC"</code>	CXC Basic / JSC 5 / SSC. 3rd JL (all spacing variants), CSEC Basic / JSC 5 / SSC / 3rd JL, code 4 (2015), 04 (2019)
<code>"CSEC General/GCE O Level"</code>	CXC Gen / GCE O (all variants), CXC GEN/GCE O LEVEL, CSEC Gen/GCE O Level, CSEC Gen,/GCE O LEVEL, code 5 (2015), 05 (2019)
<code>"NVQJ Level 1"</code>	NVQJ LEVEL 1, NVQJ Level 1, code 6 (2015), 06 (2019)
<code>"NVQJ Level 2"</code>	NVQJ LEVEL 2, NVQJ Level 2, code 7 (2015), 07 (2019)
<code>"CAPE/GCE A Level"</code>	CAPE / GCE A LEVEL (all variants), CAPE Unit 1, CAPE Unit 2/GCE A Level, CAPE UNIT 1 and 2, code 8 (2015), 09/10 (2019)
<code>"Tertiary Certificate/NVQJ Level 3"</code>	TERTIARY CERT. / DIPLOMA (pre-2012), TERTIARY CERT./DIPLOMA/ NVQJ LEVEL 3 (2016+), Tertiary Certificate / Diploma / NVQJ Level 3, code 9 (2015), 11 (2019)
<code>"Associate Degree/NVQJ Level 4"</code>	ASSOCIATE DEGREE (pre-2016), ASSOCIATE DEGREE/ NVQJ LEVEL 4, Associate degree / NVQJ Level 4, code 10 (2015), 12 (2019)
<code>"Undergraduate Degree/NVQJ Level 5"</code>	DEGREE (pre-2016), DEGREE/NVQJ LEVEL 5, Undergraduate degree/NVQJ Level 5, code 11 (2015), 13 (2019)
<code>"Higher Degree/Professional Qualification"</code>	Higher degrees and Professional Qualification (2017+), code 14 (2019)
<code>"City and Guilds"</code>	CITY AND GUILDS, City and Guilds, code 15 (2019)

Canonical String	Credentials Collapsed
"Other"	OTHER, Other, Other (specify), code 12 (2015), 20 (2019)
NA	NOT STATED, Not stated, Not Stated, 97, 98, 99, code 13 (2015), 21 (2019)

Step 2: `harmonize_exam_cat()` → `EXAM_CAT_DISAGG`

The second function is a simple one-to-one exact-match recode operating on `SCHOOL_HIGHEST_EXAM_HARMONIZED`. Because all messy variant strings have already been collapsed in Step 1, this function only needs to handle the 15 canonical strings above. Any value not in that set — including `NA` input — maps to `NA`.

`EXAM_CAT_DISAGG` category definitions

<code>EXAM_CAT_DISAGG</code> value	Input strings from <code>SCHOOL_HIGHEST_EXAM_HARMONIZED</code>	Description
"None"	"None"	No formal qualification
"Lower_Secondary"	"Junior High School Certificate", "Grade Nine Achievement Test"	Completed lower secondary (grades 7-9) without upper-secondary exam attempt
"CXC_Basic"	"CSEC Basic/JSC 5/SSC"	Sat CSEC but achieved only Basic proficiency pass. Qualifies for entry-level employment but not university matriculation
"CXC_General"	"CSEC General/GCE O Level"	Full O-Level equivalent. The standard Jamaican secondary school credential; 5+ passes including Math and English required for UWI admission
"A_Level"	"CAPE/GCE A Level"	A-Level equivalent (CAPE or legacy GCE). Required for university entrance. Note: CAPE Unit 1 only (one year of sixth form, no full A-Level) is

EXAM_CAT_DISAGG value	Input strings from SCHOOL_HIGHEST_EXAM_HARMONIZED	Description
		collapsed here for panel consistency
"Vocational_Low"	"NVQJ Level 1", "NVQJ Level 2", "City and Guilds"	Entry- and intermediate-level vocational credentials. Do not require CSEC as a prerequisite; represent a parallel track to academic secondary
"Tertiary_Cert"	"Tertiary Certificate/NVQJ Level 3"	Post-secondary certificate or diploma (community college, teacher training, nursing diploma, etc.) or NVQJ Level 3 (supervisory/technician). Requires CSEC for most programs. The survey merges these explicitly from 2016 onward
"Associate_Degree"	"Associate Degree/NVQJ Level 4"	Two-year post-secondary degree or NVQJ Level 4 (technologist). Primarily from community colleges
"Degree"	"Undergraduate Degree/NVQJ Level 5"	Bachelor's degree or NVQJ Level 5 (professional/managerial). Note: pre-2012 waves have a single undifferentiated "DEGREE" category that is mapped here rather than to "Associate_Degree", on the grounds that associate degrees were rare and the modal degree-holder in the early JSLC would have had a bachelor's
"Higher_Degree"	"Higher Degree/Professional Qualification"	Postgraduate or professional qualification. Appears only in 2017–2019 waves

EXAM_CAT_DISAGG value	Input strings from SCHOOL_HIGHEST_EXAM_HARMONIZED	Description
"Other"	"Other"	Catch-all for qualifications not elsewhere classified
NA	Any unmatched string, NA input	Not stated, missing, or unknown

Key Harmonization Decisions and Justifications

NVQJ Levels 1 and 2 in pre-2012 years

NVQJ credentials were not listed as a response option in the JSLC before 2012. This is because the NVQJ framework was still being established in Jamaica during the early 2000s. Respondents with vocational qualifications in pre-2012 waves would most likely appear in "OTHER" or "None", which flows through to EXAM_CAT_DISAGG = "Other" or "None" respectively. This creates a known limitation: the "Vocational_Low" category is structurally under-counted in early waves, and any negative trend in vocational attainment pre-to-post 2012 should be interpreted with caution.

NVQJ Levels 3, 4, 5 mapped to academic degree tiers

From 2016 onward, the JSLC explicitly merges NVQJ levels with their academic equivalents in composite categories (e.g., "TERTIARY CERT./DIPLOMA/ NVQJ LEVEL 3"). This reflects the Jamaican government's policy goal of making NVQJ credentials formally equivalent to academic ones for labor market purposes. The mapping adopted here follows the survey instrument's own logic:

- NVQJ Level 3 (supervisory/technician) → "Tertiary_Cert"
- NVQJ Level 4 (technologist) → "Associate_Degree"
- NVQJ Level 5 (professional/managerial) → "Degree"

For pre-2016 waves, "Tertiary Certificate/Diploma" maps to "Tertiary_Cert" and "Associate Degree" maps to "Associate_Degree" without NVQJ riders, since those categories were not yet merged.

CAPE Unit 1 collapsed with full CAPE/GCE A Level

In 2018, the survey separately enumerated "CAPE Unit 1" (one year of sixth form) and "CAPE Unit 2/GCE A Level" (full two-year sixth form). In earlier and later waves, these are not separated. For panel consistency, both are collapsed to "A_Level" / "CAPE/GCE A Level". This is a minor simplification: CAPE Unit 1 alone does not satisfy most university entry requirements. However,

the number of Unit 1-only respondents in 2018 is small (n=55 vs. n=23 for Unit 2), and maintaining a separate category would be undefined in all other years.

City and Guilds grouped with "Vocational_Low"

City and Guilds is a British vocational qualification occasionally offered in Jamaica, broadly equivalent to NVQJ Level 2. It appears in small numbers in 2016–2019. It is grouped with "Vocational_Low" on the basis of credential level, though it is technically a foreign qualification rather than a Jamaican one.

"Degree" in pre-2012 waves

Before 2012, the survey had a single "DEGREE" category with no associate/undergraduate distinction. This is mapped to "Degree" (undergraduate) rather than splitting across "Associate_Degree" and "Degree". The rationale is that associate degree programs were not widespread in Jamaica in the early 2000s and the modal degree-holder in those waves would have held a bachelor's degree. This decision should be acknowledged in a data appendix footnote.

Coverage by Year

Year	Has exam variable?	Format	NVQJ visible?	Notes
2000–2004	Partial	String labels	No	Simpler questionnaire; some years lack the full exam breakdown
2005	No	—	No	Variable missing in this wave
2006–2007	Yes	String labels	No	Full labels, no NVQJ
2008	Yes	String labels	No	Larger sample year (pooled waves)
2009	No	—	No	Variable missing in this wave
2010	Yes	String labels	No	
2012	Yes	String labels	Yes	NVQJ Levels 1 and 2 first appear
2013	No	—	—	Variable missing in this wave

Year	Has exam variable?	Format	NVQJ visible?	Notes
2014	Yes	String labels	Yes	Labels switch to "CSEC" from "CXC"
2015	Yes	Raw numeric	Yes	<code>as_factor()</code> not applied; codes decoded via 2012 label scheme
2016	Yes	String labels	Yes	NVQJ Levels merged with academic tiers in composite categories
2017	Yes	String labels	Yes	Higher Degree category first appears; "CAPE UNIT 1 and 2" = 0 obs
2018	Yes	String labels	Yes	CAPE Unit 1 / Unit 2 split; <code>as_factor()</code> applied
2019	Yes	Raw numeric	Yes	<code>as_factor()</code> not applied; codes decoded via 2018 label scheme (01-21)

Validation

After constructing `EXAM_CAT_DISAGG`, the cleaning script produces a validation cross-tabulation of `SCHOOL_HIGHEST_EXAM_HARMONIZED` × `EXAM_CAT_DISAGG` by year. Any string that does not match the 15 canonical strings will appear as `NA` in `EXAM_CAT_DISAGG` while retaining its original value in `SCHOOL_HIGHEST_EXAM_HARMONIZED`, making unexpected strings immediately visible. This serves as the primary QC check for the harmonization pipeline.

Variable Names Summary

Variable	Column name in <code>df_jslc_educ</code>	Source
Raw exam label (post- rename)	<code>SCHOOL_HIGHEST_EXAM_PASSED</code>	Direct from JSLC, renamed in <code>cleaning_jslc2_educationversion.R</code>
Harmonized canonical string	<code>SCHOOL_HIGHEST_EXAM_HARMONIZED</code>	Output of <code>harmonize_highest_exam()</code>

Variable	Column name in <code>df_jslc_educ</code>	Source
Disaggregated category	<code>EXAM_CAT_DISAGG</code>	Output of <code>harmonize_exam_cat()</code> , operating on <code>SCHOOL_HIGHEST_EXAM_HARMONIZED</code>

All three variables are retained in the final dataset to facilitate transparency and re-checking.

